

Serial transmission from LPC2148 to connected PC

Steps Involved

- 1) Set P0.0 pin as TxD & P0.1 pin as RxD - (USART0)
PINSEL0 = 0x00000005; // Pin Select 0
- 2) Set 8 bits - no Parity - 1 Stop bit for Txision & DLAB = 1
U0LCR = 0x83; // Line Control Register
- 3) Set Baud Rate as 9600 bps - 15MHz VPB Clock (PCLK)
U0DLL = 97; // Divisor Latch LSB
- 4) Disable Access To Divisor Latches
U0LCR = 0x03; // DLAB = 0 Bit – 7
- 5) Wait until UART0 ready to send character
while(!(U0LSR & 0x20)); // Line Status Register – THRE
- 6) Sent a character to UART Transmit Register
U0THR = 'a'; // Transmit Holding Register

Step 1 – Configure UART pins

1) Set P0.0 pin as TxD & P0.1 pin as RxD - (UART0)
`PINSEL0 = 0x00000005; // Pin Select 0`

PINSEL0 REGISTER

Bit	Symbol	Value	Function	Reset value
1:0	P0.0	00	GPIO Port 0.0	0
		01	TXD (UART0)	
		10	PWM1	
		11	Reserved	
3:2	P0.1	00	GPIO Port 0.1	0
		01	RxD (UART0)	
		10	PWM3	
17:16	P0.8	00	GPIO Port 0.8	0
		01	TXD UART1	
		10	PWM4	
		11	Reserved ^{[1][2]} or AD1.1 ^[3]	
19:18	P0.9	00	GPIO Port 0.9	0
		01	RxD (UART1)	
		10	PWM6	
		11	EINT3	

`PINSEL0 = 0x00000005; // UART 0`

`PINSEL0 = 0x00050000; // UART 1`

Step 2 – Configuring serial communication parameters

2) Set 8 bits - no Parity - 1 Stop bit for Tx & Rx & DLAB = 1
U0LCR = 0x83; // Line Control Register

Data Length – Bits for Tx or Rx

Parity Bit – (Eg. 1110 – Odd, 0101 – Even)

Stop Bits

Baud Rate – No. of bits Tx/Rx per second (bps)

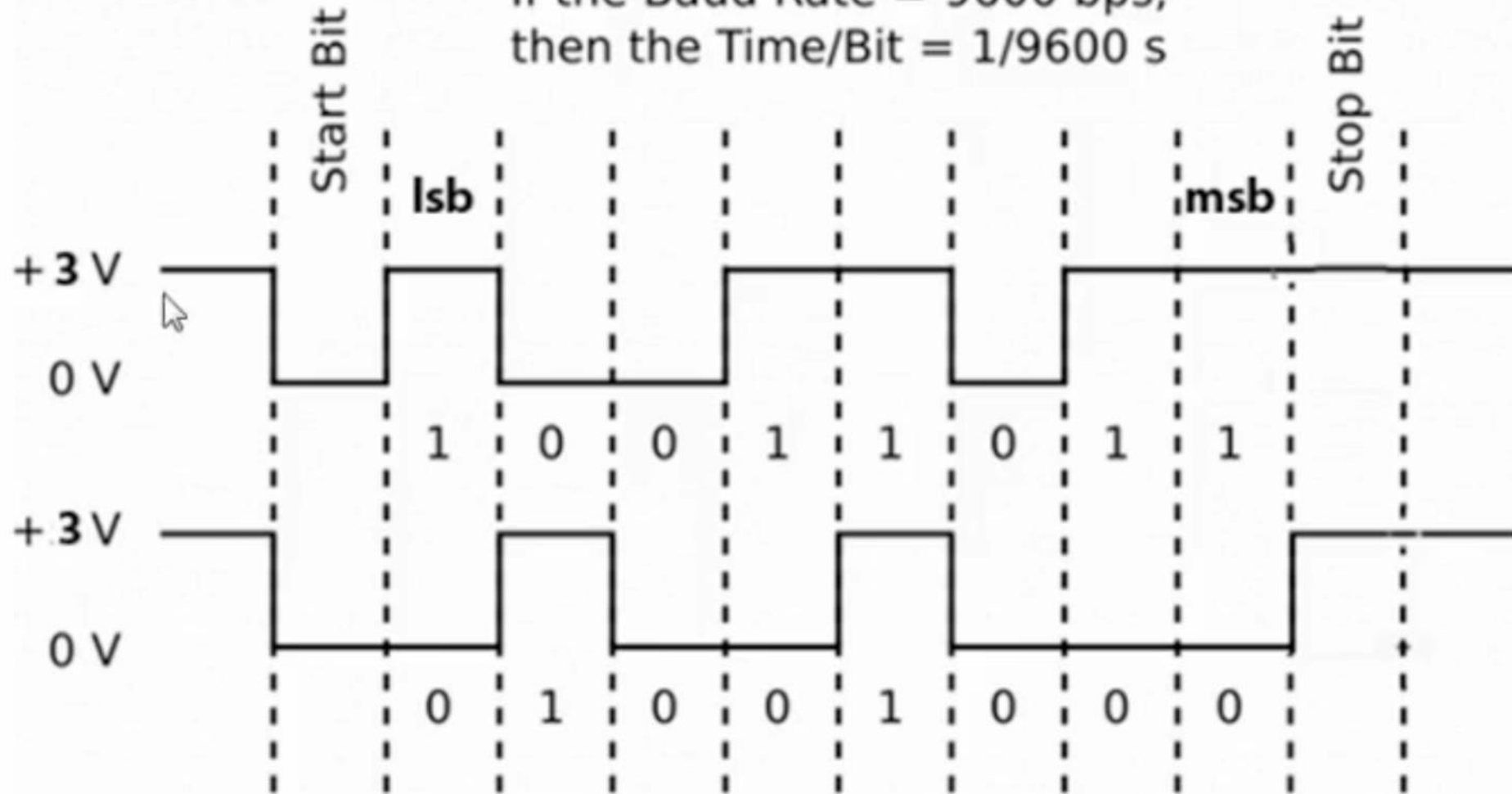
UART 0 LINE CONTROL REGISTER

Bit	Symbol	Value	Description	Reset value
1:0	Word Length Select	00	5 bit character length	0
		01	6 bit character length	
		10	7 bit character length	
		11	8 bit character length	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if U0LCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UART0 TXD is forced to logic 0 when U0LCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	

U0LCR = 0x83

1	0	0	0	0	0	1	1
8					3		

If the Baud Rate = 9600 bps,
then the Time/Bit = $1/9600$ s



3) Set Baud Rate as 9600 bps - 15MHz VPB Clock (PCLK)
U0DLL = 97; // Divisor Latch LSB

Table 99: UART0 Divisor Latch LSB register (U0DLL - address 0xE000 C000, when DLAB = 1) bit description

Bit	Symbol	Description	Reset value
7:0	DLL	The UART0 Divisor Latch LSB Register, along with the U0DLM register, determines the baud rate of the UART0.	0x01

Table 100: UART0 Divisor Latch MSB register (U0DLM - address 0xE000 C004, when DLAB = 1) bit description

Bit	Symbol	Description	Reset value
7:0	DLM	The UART0 Divisor Latch MSB Register, along with the U0DLL register, determines the baud rate of the UART0.	0x00

Baud rate calculation (generation) using Fractional Baud Rate Generator

- U0DLM: UART0 Divisor Latch MSB
- U0DLL: UART0 Divisor Latch LSB
- PCLK: Peripheral Clock
- MulVal:
- DivAddVal:

Fractional Baud Rate Generator

$$UART0_{baudrate} = \frac{PCLK}{16 \times (16 \times U0DLM + U0DLL)} \times \frac{MulVal}{(MulVal + DivAddVal)}$$

Default value of U0DLL = 0x01

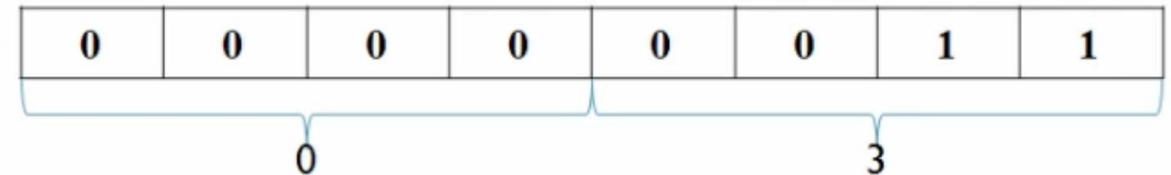
Step 4

4) Disable Access To Divisor Latches

```
U0LCR = 0x03;    // DLAB = 0 Bit - 7
```

Bit	Symbol	Value	Description	Reset value
1:0	Word Length Select	00	5 bit character length	0
		01	6 bit character length	
		10	7 bit character length	
		11	8 bit character length	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if U0LCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UART0 TXD is forced to logic 0 when U0LCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	

U0LCR = 0x03



Step 5: Check if the Transmit Holding Register is Empty

5) Wait until UART0 ready to send character

```
while((U0LSR & 0x20) == 0); // Line Status Register – THRE = 0
```

It can also be written as `while(!(U0LSR & 0x20) );`

7	6	5	4	3	2	1	0
RX FIFO Error	TEMT	THRE	BI	FE	PE	OE	RDR

Bit 5 - Transmitter Holding Register Empty

0 = U0THR has valid data

1 = U0THR empty

Bit	Symbol	Value	Description	Reset value
0	Receiver Data Ready (RDR)		U0LSR0 is set when the U0RBR holds an unread character and is cleared when the UART0 RBR FIFO is empty.	0
		0	U0RBR is empty.	
		1	U0RBR contains valid data.	
1	Overrun Error (OE)		The overrun error condition is set as soon as it occurs. An U0LSR read clears U0LSR1. U0LSR1 is set when UART0 RSR has a new character assembled and the UART0 RBR FIFO is full. In this case, the UART0 RBR FIFO will not be overwritten and the character in the UART0 RSR will be lost.	0
		0	Overrun error status is inactive.	
		1	Overrun error status is active.	
2	Parity Error (PE)		When the parity bit of a received character is in the wrong state, a parity error occurs. An U0LSR read clears U0LSR[2]. Time of parity error detection is dependent on U0FCR[0]. Note: A parity error is associated with the character at the top of the UART0 RBR FIFO.	0
		0	Parity error status is inactive.	
		1	Parity error status is active.	
5	Transmitter Holding Register Empty (THRE))		THRE is set immediately upon detection of an empty UART0 THR and is cleared on a U0THR write.	1
		0	U0THR contains valid data.	
		1	U0THR is empty.	

U0LSR & 0x20

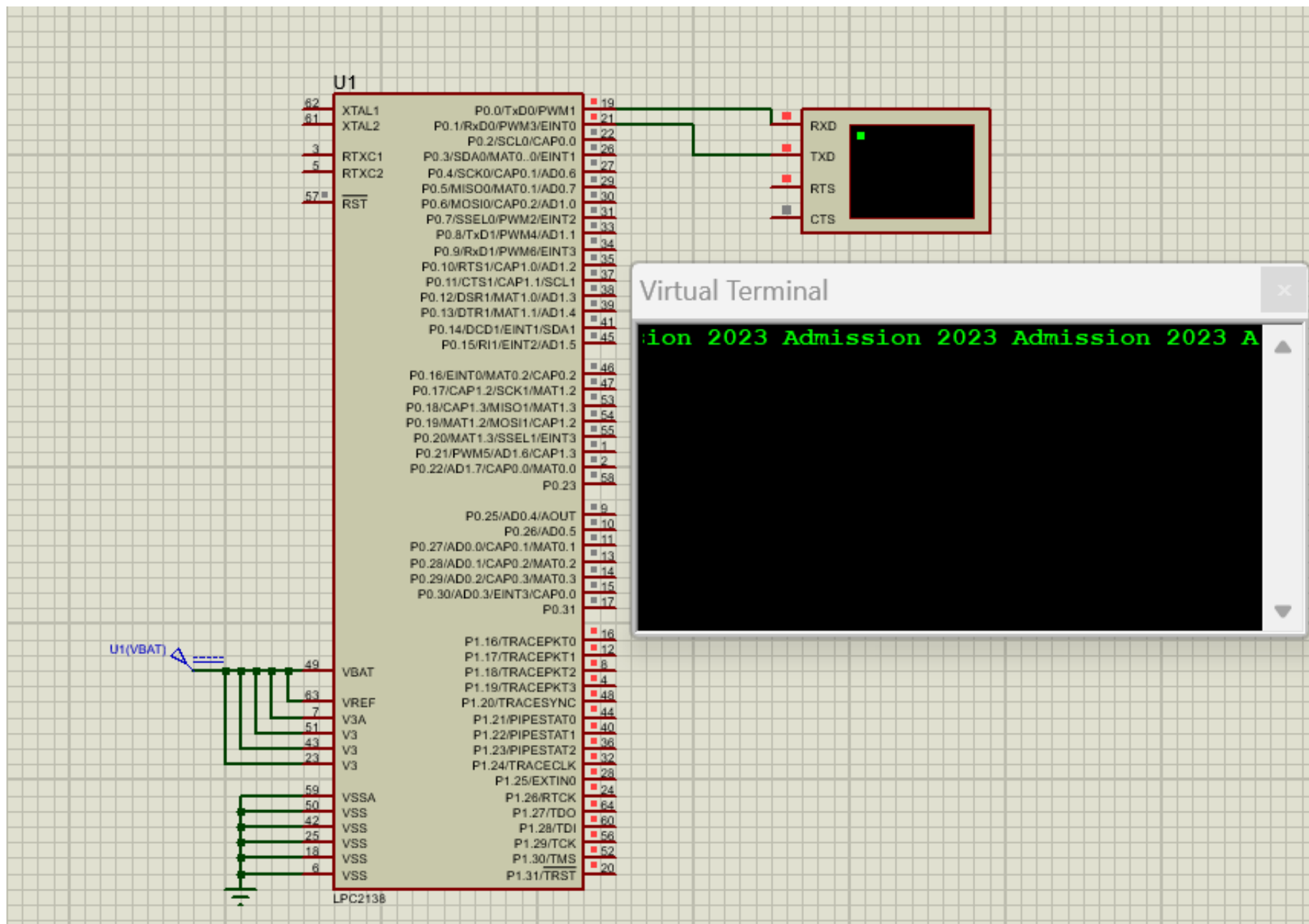
0	0	THRE	0	0	0	0	0
2/0				0			

6) Sent a character to UART Transmit Register

```
U0THR = 'a';    // Transmit Holding Register
```

```
#include <LPC214x.h>
void init(void);
void delay_ms(int);
int main()
{
    int i;
    unsigned char c[] = "Admission 2023 \n"; //Send data by writing to U0THR
    init();
    while(1)
    {
        for(i=0;i<=17;i++)
        {
            U0THR = c[i];
            while(!(U0LSR & 0x20)); // Check status of one bit of U0LSR
            delay_ms(250);
        }
    }
}
```

```
void init()
{
    PINSEL0 = 0x05;
    UOLCR = 0x83; //8-N-1, enable divisors
    UODLL = 0x62; //9600 baud rate
    UODLM = 0x00;
    UOLCR = 0x03; //8-N-1, disable divisors
}
void delay_ms(int x)
{
    int a,b;
    for(a=0;a<x;a++)
    {
        for(b=0;b<3000;b++);
    }
}
```

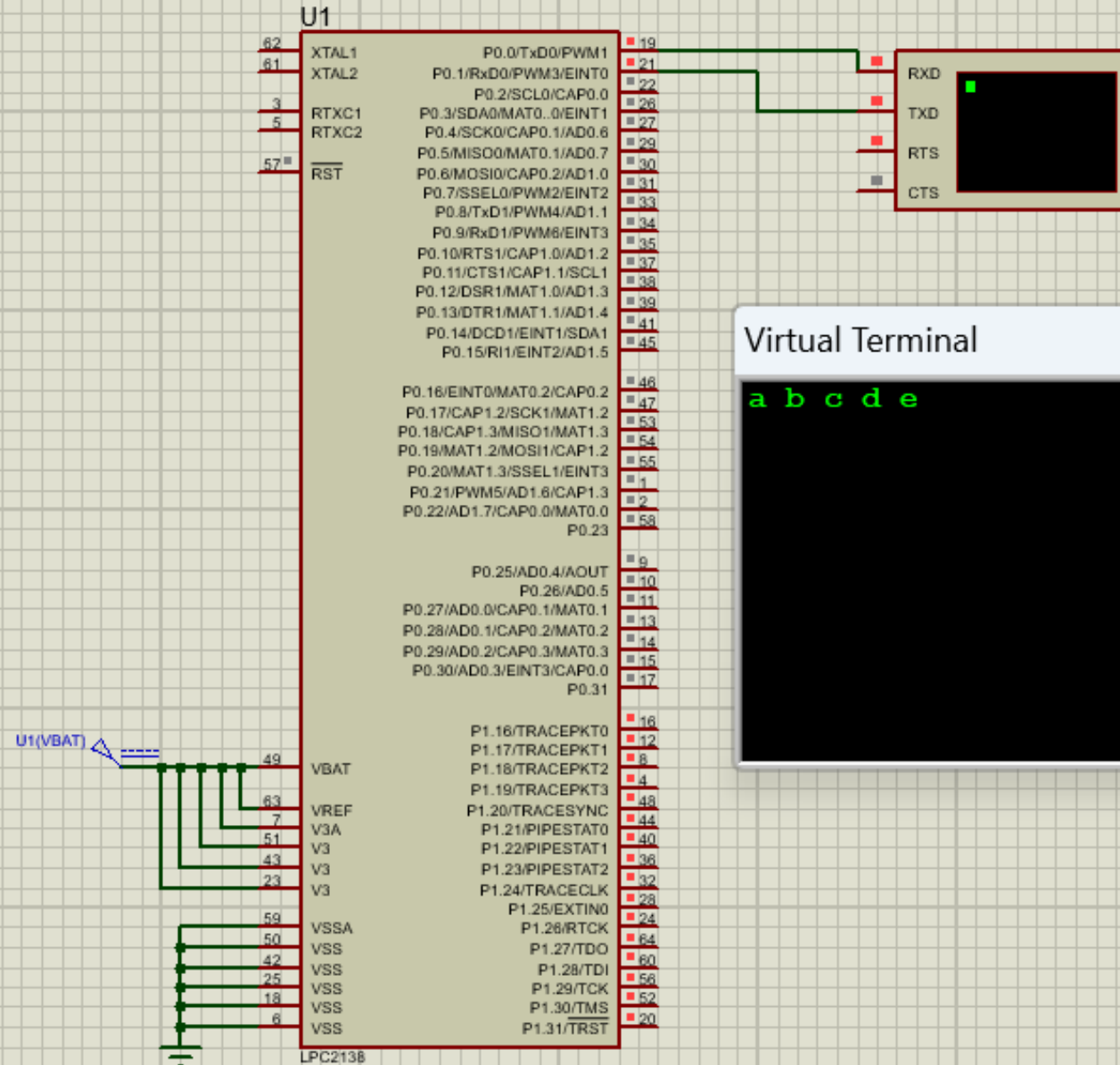
Summary of key registers used

Register	Purpose
PINSEL0	Select UART function for pins
U0LCR	Configure UART frame format
U0DLL / U0DLM	Set baud rate
U0LSR	Check transmission/reception status
U0THR	Send character
U0RBR	Receive character

EXERCISE:

Write a program for the serial reception of the data from the virtual terminal to LPC2148.

```
#include <LPC214x.h>
void init(void);
int main()
{
    unsigned char data;
    init();
    while(1)
    {
        while(!(U0LSR & 0x01));
        data=U0RBR;
        while(!(U0LSR & 0x20)); // Check status of one bit of U0LSR
        U0THR = data;
    }
}
void init()
{
    PINSEL0 = 0x05;
    U0LCR = 0x83; //8-N-1, enable divisors
    U0DLL = 0x62; //9600 baud rate
    U0DLM = 0x00;
    U0LCR = 0x03; //8-N-1, disable divisors
}
```



Virtual Terminal

a b c d e